Binary-field multiplicative-divisor eligibility

Derived field screen, 2026-10-17 UTC. Exact factorizations of $2^n - 1$ were checked at $n = 50, 53, 63$. The chart shows the nearest multiplicative divisor to twice the nominal default factor-base size across $m = 2…8$. This constructs no binary curve or base and measures no index-calculus cost.

Nearest divisor of 2^n−1 vs 2 × nominal base size

	m=2	m=3	m=4	m=5	m=6	m=7	m=8
GF(2^50)	0.0%	0.5%	12.7%	18.8%	12.6%	21.4%	4.9%
GF(2^53)	78.5%	76.9%	64.9%	80.6%	99.9%	99.8%	99.7%
GF(2^63)	23.5%	3.3%	9.6%	17.3%	7.4%	43.3%	9.5%

Green: ≤15%; amber: 15–25%; red: >25%. Rounded to 0.1%.

Source: research/icperf_binary_divisor_20261017/field_fit.json. Field eligibility only.

Field	Cells with divisor within 15% of $2B$
GF(2^50)	m=2,3,4,6,8
GF(2^53)	None of m=2…8
GF(2^63)	m=3,4,6,8

For $n = 63$, $m = 8$, the nominal B is 406, the closest divisor is 889, and the integer gap is $\frac{77}{812} = 9.48\%$. For $n = 53$, the field has only eight divisors and none fits. The default-size changes over the full Hasse interval are at most one and leave all 21 classifications unchanged.

Remaining measurement

A concrete certified ordinary-binary or Koblitz curve, actual target subgroup, point lifts, realized base, matched controls and all collection, sparse-solve and target costs are needed. The prime-field constructor cannot be used for binary-field arithmetic. Koblitz signed-Frobenius bases are a distinct path. The canonical ECDLP frontier graph remains unchanged because this screen adds no reviewed cost comparison.

Machine-readable data: research/icperf_binary_divisor_20261017/field_fit.json (SHA-256 a5d2387e5cebdf34c3a66e766ab14e7d43878c24998132eb32c2d19aa6b8c6b9).